

Industrial Internship Report

on

“CMS Blog Platform”

Prepared by

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Organization	UniConverge Technologies Pvt Ltd (UCT)
Platform	upskill Campus & The IoT Academy
Project	CMS Blog Platform
Duration	6 Weeks
Submitted By	Anish Kumar Singh

Executive Summary

This report documents the Industrial Internship undertaken through upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT). The internship spanned 6 weeks and was centered around the development of a CMS Blog Platform — a fully functional content management system that enables users to create, manage, publish, and organize blog content through an intuitive web interface. The platform was designed with role-based access, a rich text editor, SEO-friendly URL routing, and a REST API backend. This internship provided valuable exposure to real-world software engineering workflows, including requirement analysis, system design, implementation, testing, and documentation. The experience significantly enhanced technical and professional skills and offered deep insight into industry-grade web development practices.

Table of Contents

1	Preface	4
2	Introduction	5
2.1	About UniConverge Technologies Pvt Ltd	5
2.2	About upskill Campus	6
2.3	Objective	7
2.4	Reference	7
2.5	Glossary	8
3	Problem Statement	9
4	Existing and Proposed Solution	10
5	Proposed Design / Model	11
5.1	High Level Diagram	11
5.2	Low Level Diagram	12
5.3	Interfaces	12
6	Performance Test	13
6.1	Test Plan / Test Cases	13
6.2	Test Procedure	14
6.3	Performance Outcome	14
7	My Learnings	15
8	Future Work Scope	16

1. Preface

Over the course of six weeks, I worked on developing a Content Management System (CMS) Blog Platform as part of my industrial internship with UniConverge Technologies Pvt Ltd (UCT), facilitated by upskill Campus and The IoT Academy. This experience was transformative both technically and professionally.

The CMS Blog Platform project gave me hands-on exposure to the full software development lifecycle — from gathering requirements and designing architecture to coding, testing, and delivering a functional product. Working within an industry-relevant timeline sharpened my problem-solving abilities and taught me the importance of structured planning.

I am deeply grateful to the upskill Campus team, UCT mentors, and my peers for their continuous support and guidance. This internship has reinforced my passion for web development and strengthened my confidence to tackle real-world engineering challenges. I encourage my peers and juniors to seize such opportunities, as there is no substitute for applied learning.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) was established in 2013 and operates in the Digital Transformation domain, providing industrial solutions with a prime focus on sustainability and Return on Investment (RoI).

UCT leverages cutting-edge technologies including:

- Internet of Things (IoT) and Cyber Security
- Cloud Computing (AWS, Azure)
- Machine Learning and AI
- Communication Technologies (4G/5G/LoRaWAN)
- Java Full Stack, Python, and Front-End Frameworks

Key UCT products include:

UCT Insight — IoT Platform:

An IoT platform designed for rapid deployment of IoT applications, built with Java backend and ReactJS frontend, supporting MySQL and NoSQL databases. Features include device connectivity via MQTT, CoAP, HTTP, Modbus TCP, and OPC UA; cloud and on-premises deployments; custom dashboard building; analytics and reporting; alert and notification systems; and integration with third-party applications (Power BI, SAP, ERP).

Smart Factory Platform (FactoryWatch):

A platform designed for smart factory monitoring needs, offering scalable production and asset monitoring, OEE and predictive maintenance, digital twin capabilities, and a modular SaaS architecture.

2.2 About upskill Campus (USC)

upskill Campus, along with The IoT Academy and in association with UCT, facilitates the smooth execution of the complete internship process. USC is a career development platform that delivers personalized executive coaching in an affordable, scalable, and measurable way. USC aims to upskill 1 million learners over the next five years.

Website: <https://www.upskillcampus.com/>

2.3 Objectives of this Internship Program

- Gain practical experience of working in the industry.
- Solve real-world problems through software design and development.
- Improve job prospects through project-based learning.
- Develop a deeper understanding of web technologies and CMS architecture.
- Foster personal growth including communication and problem-solving skills.

2.4 Reference

[1] WordPress CMS Documentation — <https://wordpress.org/documentation/>

[2] Ghost CMS Official Guide — <https://ghost.org/docs/>

[3] Node.js and Express.js Documentation — <https://nodejs.org/en/docs/>

2.5 Glossary

Terms	Acronym
Content Management System	CMS
Application Programming Interface	API
Search Engine Optimization	SEO
User Interface	UI
User Experience	UX
Representational State Transfer	REST
Structured Query Language	SQL

3. Problem Statement

In the modern digital era, individuals and organizations require an easy-to-use platform for publishing and managing blog content without deep technical knowledge. Existing solutions are either too complex for beginners or lack the customization needed for professional use.

The assigned problem statement required designing and developing a CMS Blog Platform that:

- Allows multiple user roles (Admin, Editor, Author, Viewer) to interact with content differently.
- Provides a rich text editor for creating and formatting blog posts.
- Supports media uploads (images, files) attached to blog entries.
- Generates SEO-friendly URLs and meta-tag support for each post.
- Delivers a responsive frontend accessible across desktop and mobile devices.
- Exposes a REST API for content retrieval by third-party applications.

4. Existing and Proposed Solution

Existing Solutions & Their Limitations

Several CMS platforms already exist in the market:

- WordPress: Feature-rich but heavy, plugin-dependent, and often overkill for lightweight blogging.
- Ghost: Clean and fast but limited in multi-user collaboration and role customization.
- Blogger: Simple but severely limited in customization and lacks a modern API.

Common limitations include tight coupling of frontend and backend, limited REST API support, and difficulty in extending role-based access control.

Proposed Solution

The proposed CMS Blog Platform is a decoupled (headless) web application built with a Node.js/Express.js backend exposing a RESTful API, and a React.js frontend for the admin dashboard and public blog view.

Value additions include:

- JWT-based authentication with role-based access control (RBAC).
- Integrated rich text editor (Quill.js) for formatting blog posts.
- Auto-generated SEO slugs and customizable meta tags per post.
- Cloud image storage integration (Cloudinary).
- Responsive UI with Tailwind CSS for mobile-first design.

Code Repository (GitHub): <https://github.com/anishkumarsingh/cms-blog-platform>

Report Repository (GitHub): <https://github.com/anishkumarsingh/cms-blog-platform-report>

5. Proposed Design / Model

The system follows a three-tier architecture: Presentation Layer (React.js), Business Logic Layer (Express.js), and Data Layer (MongoDB).

5.1 High Level Diagram

[Figure 1: High Level Architecture Diagram of the CMS Blog Platform]

The high-level diagram illustrates the interaction between:

- Users (Admin, Editor, Author, Viewer) accessing the React frontend.
- React frontend communicating with the Express.js REST API via HTTP/HTTPS.
- Express.js handling business logic and authentication (JWT).
- MongoDB database storing users, posts, categories, and media metadata.
- Cloudinary for image storage and delivery.

5.2 Low Level Diagram

[Figure 2: Low Level Diagram — Database Schema & API Endpoint Mapping]

The low-level design covers:

- Database collections: Users, Posts, Comments, Categories, Tags, Media.
- REST API routes: /api/posts, /api/users, /api/auth, /api/media.
- Middleware stack: Auth middleware, validation middleware, error handler.

5.3 Interfaces

Key interfaces include:

- Admin Dashboard: Post management, user management, category editor, analytics overview.
- Post Editor: Quill.js rich text editor with image upload, tag selection, and SEO fields.
- Public Blog View: Responsive card-based layout, search, category filter, paginated listing.
- REST API: JSON responses with JWT-secured endpoints for CRUD operations on posts.

6. Performance Test

Performance testing was conducted to ensure the platform meets industry-grade responsiveness and reliability standards.

6.1 Test Plan / Test Cases

Test ID	Test Case	Expected Result	Status
TC-01	Login with valid credentials	Redirect to dashboard	PASS
TC-02	Create and publish a blog post	Post visible on public blog	PASS
TC-03	Upload image in post editor	Image stored on Cloudinary	PASS
TC-04	API endpoint GET /api/posts	Returns paginated JSON	PASS
TC-05	Unauthorized access to admin routes	401 Unauthorized response	PASS
TC-06	Search posts by keyword	Matching posts returned	PASS

6.2 Test Procedure

Testing was conducted in a local development environment using Postman for API testing and Jest for unit tests. End-to-end user flows were tested manually in Chrome and Firefox browsers.

6.3 Performance Outcome

- Average API response time: ~120ms for GET requests.
- Page load time (public blog): ~1.4 seconds on 4G network.
- Lighthouse performance score: 87/100.
- All 6 test cases passed successfully.
- Zero critical security vulnerabilities detected via OWASP ZAP scan.

7. My Learnings

This internship provided an exceptional opportunity to bridge the gap between academic knowledge and industrial application. Key learnings include:

- Hands-on experience building a full-stack web application using Node.js, Express.js, React.js, and MongoDB.
- Understanding of RESTful API design principles, JWT authentication, and RBAC implementation.
- Exposure to cloud services including Cloudinary for media management.
- Appreciation for software documentation — clear README files, API docs, and code comments.
- Improved ability to manage time, prioritize tasks, and meet deadlines within a 6-week sprint.
- Collaboration and communication skills through weekly review meetings with UCT mentors.

These learnings will directly contribute to my growth as a software engineer and equip me for professional roles in full-stack development, product engineering, and cloud-based application development.

8. Future Work Scope

Several enhancements can be made to the CMS Blog Platform in the future:

- **AI-powered Content Suggestions:** Integrate an LLM-based assistant to help authors generate post outlines and improve SEO scores.
- **Multi-language Support:** Add internationalization (i18n) to support regional language blogs.
- **Analytics Dashboard:** Embed page view tracking, reader demographics, and post performance metrics.
- **Email Newsletter Integration:** Allow subscribers to receive new posts via automated email campaigns.
- **PWA Support:** Convert the public blog into a Progressive Web App for offline reading.
- **Kubernetes Deployment:** Containerize the platform with Docker and orchestrate with Kubernetes for enterprise-scale deployment.

These features, while beyond the current 6-week scope, represent a clear roadmap for evolving this platform into a production-grade CMS solution.